Scalability Plan:

Script-Controlled ACL – Restrict Record Access Based on Field Value

Phase 8-Scalability & Future Plan

Date	30 September 2026	**Team ID**	6ab3ed08a238dc999eb3fda5
Project Name	Script-Controlled ACL - Restrict Record Access Based on Field Value	**Maximum Marks**	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution (Script-Controlled Access Control Lists - ACL in ServiceNow) can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Hardcoded Role & Branch Evaluation in ACL Script	Restricts flexibility; adding new branches (e.g., ECE, CSE) requires manual script updates in ACL.	High
2	Manual User & Role Assignment	Assigning custom roles (bb1, bb2, bb3, bb4) manually to individual users creates operational overhead at scale.	Medium
3	Single Custom Table Scope (u_institution_details)	Access control is isolated to a single table without cross-table or dynamic security rule inheritance.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Manual role assignment for EEE User and test accounts.	Dynamic Group-Based Role Assignment (assign roles via Department Groups like 'EEE Dept').
2	Data Storage	Static records in u_institution_details table.	Automated data imports via Transform Maps / IntegrationHub for multi-department scale.
3	Performance	Scripted ACL evaluates gs.hasRole() per record query.	Optimize ACL scripts using System Properties / Dynamic Conditions to minimize DB query overhead.
4	Security	READ, CREATE, WRITE, DELETE ACLs with static roles.	Implement Attribute-Based Access Control (ABAC) and Security Attribute Conditions for multi-tenancy.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Dynamic Branch-Based ACL Engine (Dynamic Scripting for ECE, CSE, EEE)	Q3 2024	Eliminates hardcoded roles; scales security rules dynamically for all institution branches.
Phase 3	Service Portal & Mobile App Integration with Field-Level Access Control	Q4 2024	Enforces secure record viewing and field obfuscation across Service Portal and Mobile interfaces.
Phase 4	Automated Security Audit Logging & Compliance Dashboards	Q1 2025	Provides real-time visibility into ACL access evaluations, denied attempts, and role management.